ACML101
Physical Chemistry Tutorial III

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Reversible Reaction

Q1. For an opposite reaction

$$A \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} B$$

given k_1 = 0.006 min^{-1} and k_{-1} = 0.002 min^{-1}, if only A exists at the beginning with a concentration of 1 $mol.dm^{-3}$, determine:

(1) The time needed when concentrations of A and B are equal.

(2) Concentrations of A and B after 100 min.

Ans: (1) 137 min (2) [A]=0.587 mol L^{-1}, [B]=0.413 mol L^{-1}

Ans:.

For a reversible reaction

$$\ln \frac{[A]_0 - [A]_{eq}}{[A] - [A]_{eq}} = (k_1 + k_{-1})t \quad \text{--(i)}$$

$$\frac{[B]_{eq}}{[A]_{eq}} = \frac{k_1}{k_{-1}} = 3 \;; \quad [B]_{eq} = 3[A]_{eq}$$

$$[A]_0 = [A] + [B] = [A]_{eq} + [B]_{eq} = 1 \text{ mol}L^{-1}$$

$$[A]_{eq} + [B]_{eq} = 1$$

$$[A]_{eq} + 3[A]_{eq} = 1$$

$$[A]_{eq} = 0.25$$

(1) $[A] = [B]$

$[A] + [B] = [A]_0 \;; \quad 2[A] = [A]_0$

$[A] = \frac{[A]_0}{2} = 0.5$

From (i)

$$\ln \frac{1 - 0.25}{0.5 - 0.25} = (k_1 + k_{-1})t$$

$$\ln 3 = 0.008 \times t$$

$$t = 137 \text{ min}$$

(2)

$$\ln \frac{1-1/4}{x-\frac{1}{4}} = 0.008 \times 100$$

$$= 0.8$$

$$\ln \frac{3/4}{\frac{4x-1}{4}} = 0.8$$

$$\ln \frac{3}{4x-1} = 0.8$$

$$\frac{3}{4x-1} = 2.22$$

$$4x-1 = \frac{3}{2.22} = 1.35$$

$$4x = 2.35$$

$$x = 0.5875$$

$$[A] = 0.5875 \text{ mol L}^{-1}$$

$$[A] + [B] = 1 \text{ mol L}^{-1}$$

$$[B] = 1 - 0.5875$$

$$= 0.413 \text{ mol L}^{-1}$$

Q2. The reaction of β–glucose changing to α–glucose

$$\beta\text{-}C_6H_{12}C_6 \rightleftharpoons \alpha\text{–}C_6H_{12}C_6$$

is a first order opposite reaction with k_1+k_{-1}=0.0016 min^{-1} and an equilibrium constant K_c = 0.557. Calculate k_1 and k_{-1}.

Ans: k_1=0.000572 min^{-1}, k_{-1}=0.00102 min^{-1}

Ans:-

$$k_1 + k_{-1} = 0.0016$$

$$K_c = \frac{k_1}{k_{-1}} = 0.557$$

$$0.557\, k_{-1} + k_{-1} = 0.0016$$

$$k_{-1} = 1.02 \times 10^{-3}$$

$$k_1 = 5.72 \times 10^{-4}$$

CONSECUTIVE REACTION AND SSA

Q3. The reaction of hydrolysis of L –2, 3 – ; 4, 6 – diacetone guluronic acid ($C_{12}H_{18}O_6$) in acid solution forming ascorbic acid ($C_6H_8O_6$, Vitamin C),

$$C_{12}H_{18}O_6 \xrightarrow{k_1} C_6H_8O_6 \xrightarrow{k_2} \text{dissociation products}$$

is a first order consecutive reaction with k_1= 4.2 x 10^{-3} min^{-1}, k_2 = 0.020 x 10^{-3} min^{-1} at 50 ºC and k_1 = 410 x 10^{-3} min^{-1}, k_2 = 0.090 x 10^{-3} min^{-1} at 60 ºC. Determine the most favourable reaction time and the corresponding maximum yield for the production of ascorbic acid.

Ans:

Ans.

For Consecutive Reaction

$$[I] = \frac{k_a}{k_a - k_b}\left[\exp(-k_a t) - \exp(-k_b t)\right][A]_0$$

[I] will be maximum at time t when

$$\frac{d[I]}{dt} = 0$$

$$\frac{k_a[A]_0}{k_a - k_b}\left[-k_a \exp(-k_a t) + k_b \exp(-k_b t)\right] = 0$$

$$k_b \exp(-k_b t) = k_a \exp(-k_a t)$$

$$\exp(k_a - k_b)t = \frac{k_a}{k_b}$$

$$t_{max} = \frac{\ln k_a/k_b}{k_a - k_b}$$

$k_1 >> k_2$

$$[I] = [A]_0 \exp(-k_2 t)$$

At 50 °c

$$t_{max} = \ln \frac{4.2\times10^{-3}/0.02\times10^{-3}}{(4.2 - 0.02)\times10^{-3}}$$

$$= 1279.2 \ \%.$$

$$[I]_{max} = [A]_0 \exp(-k_2 t_{max})$$

$$\frac{[I]_{max}}{[A]_0} \times 100 = 97.5\ \%.$$

At 60 °c,

$$t_{max} = 20.5 \text{ min}$$

$$\frac{[I]_{max}}{[A]_0} \times 100 = 99\ \%.$$

Q4. The following is a highly simplified mechanism for the decomposition of C_3H_8:

$$C_3H_8 \xrightarrow{k_1} CH_3^{\cdot} + C_2H_5^{\cdot}$$
$$CH_3^{\cdot} + C_3H_8 \xrightarrow{k_2} CH_4 + C_3H_7^{\cdot}$$
$$C_3H_7^{\cdot} \xrightarrow{k_3} C_2H_4 + CH_3^{\cdot}$$
$$CH_3^{\cdot} + CH_3^{\cdot} \xrightarrow{k_4} C_2H_6$$

Pick out the intermediates and formulate a steady state expression, where possible. From these, find expressions for each steady state concentration, and then formulate the overall rate of reaction in terms of the rate of production of CH_4.

ns: -

$$\frac{d[CH_4]}{dt} = k_2[CH_3][C_3H_8] \quad \text{— (i)}$$

Applying SSA to $[CH_3]$

$$\frac{d[CH_3]}{dt} = k_1[C_3H_8] - k_2[CH_3][C_3H_8] + k_3[C_3H_7] - 2k_4[CH_3]^2 \simeq 0 \quad \text{— (ii)}$$

Similarly, Applying SSA to C_3H_7

$$\frac{d[C_3H_7]}{dt} = k_2[CH_3][C_3H_8] - k_3[C_3H_7] = 0 \quad \text{— (iii)}$$

$$k_2[CH_3][C_3H_8] = k_3[C_3H_7] \quad \text{— (iv)}$$

From (ii) and (iv)

$$k_1[C_3H_8] - \cancel{k_2[CH_3][C_3H_8]} + \cancel{k_2[CH_3][C_3H_8]} - 2k_4[CH_3]^2 = 0$$

$$2k_4[CH_3]^2 = k_1[C_3H_8]$$

$$[CH_3] = \left(\frac{k_1}{2k_4}\right)^{1/2}[C_3H_8]^{1/2}$$

From (i)

$$\frac{d[CH_4]}{dt} = \left(\frac{k_1}{2k_4}\right)^{1/2} \times k_2[C_3H_8]^{3/2}$$

Parallel Reaction

Q5. For a parallel reaction of chlorobenzene and chlorine in a CS_2 solution,

$$C_6H_5Cl + Cl_2 \longrightarrow HCl + o\text{-}C_6H_4Cl_2, \quad C_6H_5Cl + Cl_2 \longrightarrow HCl + p\text{-}C_6H_4Cl_2,$$

in the presence of iodine as a catalyst, with the same initial concentrations of 0.5 mol.dm^{-3} for C_6H_5Cl and Cl_2 at a constant temperature and a constant concentration of iodine, after 30 min, 15 % of C_6H_5Cl transforms to *o*-$C_6H_4Cl_2$, 25 % to *p*-$C_6H_4Cl_2$. Calculate the second order rate coefficients k_1 and k_2 for the two reactions.

Ans: k_1= 0.0166 L mol^{-1} min^{-1}, k_2=0.0278 Lmol^{-1} min^{-1}

Ans:-

$$A + B \xrightarrow{k_1} C + E$$
$$A + B \xrightarrow{k_2} D + E$$

$A = C_6H_5Cl$, $B = Cl_2$ $C = o\text{-}C_6H_4Cl_2$

$D = p\text{-}C_6H_4Cl_2$, $E = HCl$

$$-\frac{d[A]}{dt} = k_1[A][B] + k_2[A][B]$$

$$= (k_1+k_2)[A][B]$$

since $[A] = [B]$

$$-\frac{d[A]}{dt} = (k_1+k_2)[A]^2$$

$$\frac{1}{[A]} - \frac{1}{[A]_0} = (k_1+k_2)t$$

$$\frac{[A]_0}{[A]} - 1 = (k_1+k_2)[A]_0 t \quad \text{—(i)}$$

$$[A] = [A]_0 - [A]_0 \times \frac{15}{100} - [A]_0 \times \frac{25}{100}$$

$$= [A]_0(1 - 0.15 - 0.25) = [A]_0 \times 0.6$$

$$\frac{[A]_0}{[A]} = \frac{1}{0.6} \quad \text{—(ii)}$$

From (i)

$$\frac{1}{0.6} - 1 = (k_1+k_2) \times 30 \times 0.5$$

$$k_1 + k_2 = 0.0444$$

$$\frac{k_1}{k_2} = \frac{15}{25} = \frac{3}{5}$$

$$k_1 + \frac{5}{3} k_1 = 0.0444$$

$$\frac{8}{3} k_1 = 0.0444$$

$$k_1 = 0.0167 \quad L\ mol^{-1}\ min^{-1}$$

$$k_2 = \frac{5}{3} k_1 = 0.278 \quad L\ mol^{-1}\ min^{-1}$$

Temperature dependence of the rate constant

(6) Consider a reaction $A(g) \rightarrow B(g)$ taking place in two elementary steps in presence of a catalyst M

$$A(g) + M(g) \rightleftharpoons A(g)^* + M(g)$$

$$A(g)^* \rightarrow B(g)$$

Calculate k_{obs} in terms of rate constants (k_1, k_{-1} and k_2). If $k_{-1}[M] \gg k_2$, calculate observed activation energy in terms activation energy (threshold energy) of individual steps.

Ans

$$A(g) + M(g) \underset{k_{-1}}{\overset{k_1}{\rightleftharpoons}} A(g)^* + M(g)$$

$$A(g)^* \xrightarrow[k_2]{} B(g)$$

$$\frac{d[B]}{dt} = k_2 \; [A(g)]^*$$

$$\frac{d[A]^*}{dt} = k_1 [A][M] - k_{-1} [A]^*[M] - k_2 [A^*]$$

Applying SSA to $[A]^*$

$$k_1 [A][M] - k_{-1} [A^*][M] - k_2 [A]^* = 0$$

$$[A^*] = \frac{k_1 [A][M]}{k_{-1}[M] + k_2}$$

$$\frac{d[B]}{dt} = \frac{k_1 k_2 [A][M]}{k_{-1}[M] + k_2}$$

$$k_{-1}[M] >> k_2$$

$$\frac{d[B]}{dt} = \frac{k_1 k_2 [A][M]}{k_{-1}[M]} = \frac{k_1 k_2}{k_{-1}} [A]$$

$$k_{eff} = \frac{k_1 k_2}{k_{-1}}$$

$$\ln k_{eff} = \ln k_1 + \ln k_2 - \ln k_{-1}$$

$$\frac{E_{a,eff}}{RT^2} = \frac{E_1}{RT^2} + \frac{E_2}{RT^2} - \frac{E_{-1}}{RT^2}$$

$$E_{a,eff} = E_1 + E_2 - E_{-1}$$

Mechanism of the reaction

Q7. Suggest one reaction scheme for the oxidation of vanadium (III) ions by mercury (II) ions on the basis of rate equation given. Please provide the logic behind the proposed mechanism.

$$2V^{3+} + +2Hg^{2+} = 2V^{IV} + Hg_2^{2+}$$

$$-\frac{d[V^{3+}]}{dt} = \frac{k[V^{3+}]^2[Hg^{2+}]}{k'[V^{IV}]+[V^{3+}]}$$

Ans:- $2V^{3+} + 2Hg^{2+} \rightarrow 2V^{4+} + Hg_2^{2+}$

$$-\frac{d[V^{3+}]}{dt} = \frac{k[V^{3+}]^2[Hg^{2+}]}{k'[V^{4+}]+[V^{3+}]}$$

~~The~~ Two terms in denomination

$\Rightarrow$ At least 2 steps with first step reversible

$\rightleftharpoons$

$\longrightarrow$

V^{4+} is the product in denominator

And V^{3+} is reactant in denominator.

$\rightleftharpoons I + V^{4+}$

$I + V^{3+} \longrightarrow$

The stoichiometry of the second step Can be ~~refer~~ known by neglecting the term V^{3+} in denominator

$$v = \frac{k[V^{3+}]^2[Hg^{2+}]}{k'[V^{4+}]}$$

Stoichiometry of reactant in second step

$= 2V + 1Hg - 1V = 1V + 1Hg$

Charge $= 3\times 2 + 2 - 4 = 8 - 4 = 4$

V^{3+} is already a reactant in the second step.

So stoichiometry of intermediate

$= 1\ Hg$

Charge $= 4 - 3 = 1$

$$Hg^{2+} + V^{3+} \rightleftharpoons Hg^{+} + V^{4+}$$

$$Hg^{+} + V^{3+} \rightarrow Hg + V^{4+}$$

The reactant and product in the first step and second step respectively can be determined ~~based on~~ based on information on reactant and keeping the balance of Equation in mind.

Since Hg is not a product, it must be utilized in the next step such that product Hg_2^{2+} is formed.

$$Hg^{2+} + V^{3+} \rightleftharpoons Hg^{+} + V^{4+}$$

$$Hg^{+} + V^{3+} \rightarrow Hg + V^{4+}$$

$$Hg + Hg^{2+} \rightarrow Hg_2^{2+}$$

$$2V^{3+} + 2Hg^{2+} \rightarrow 2V^{4+} + Hg_2^{2+}$$